



R V COLLEGE OF ENGINEERING
(An autonomous institution affiliated to VTU, Belgaum)
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

IM233AI – WORK SYSTEMS DESIGN

UNIT I- Part B

Topic ID 1 Introduction to Work Systems and Productivity

Q No	Question	Marks	Difficulty Level	BT Level	COs
1.	Explain the pyramidal structure of work as described by Groover. How does it aid in work system design?	08	M	3	1
2	Discuss the importance of time in work systems. How is time used as a basis for productivity measurement?	08	M	3	1
3	Define physical work systems. Describe their components with examples.	08	M	2	1
4	How is work system a field of professional practice in industrial engineering? Illustrate with real-world applications.	08	M	3	1
5	Differentiate between manual work systems and worker-machine systems. Provide examples and implications for productivity.	08	M	3	1
6	Describe the role of standard time in work measurement. How is it established and used in planning?	08	M	2	3
7	Explain the concept of productivity improvement through work system redesign. Provide two techniques with examples.	08	M	3	1
8	Analyze the three categories of excess non-productive activities generally carried out in organizations.	08	M	4	1
9	Distinguish between Production and Productivity. Also justify the rationale for considering labor productivity as a measure of organizational efficiency and effectiveness.	08	M	3	1
10	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/day for 22 days in the month. (a) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? (b) Suppose that in the next month, the same work group produced 600 units but there were only 20 workdays in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	08	M	3	1



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11	A work group of 10 workers in a certain month produced 7200 units of output working 8 hr/day for 22 days in the month. Determine the labor productivity ratio using: (a) Units of output per worker-hour and (b) Units of output per worker-month. (c) Suppose that in the next month, the same work group produced 6800 units but, there were only 20 workdays in the month. For each productivity measure in (a) and (b), determine the productivity index for the next month using the prior month as a base.	08	M	3	1
12	A work group of 20 workers in a certain month produced 8600 units of output working 8 hr/day for 21 days.	08	M	3	1

Topic ID 2 Manual Work and Worker-Machine Systems

13	Define a manual work system. Explain its key components and give two examples.	08	L	2	1
14	Compare manual work systems and worker-machine systems in terms of productivity, ergonomics, and cost.	08	M	4	1
15	Describe the elements of a worker-machine system. How does it differ from a fully automated system?	08	M	2	1
16	Describe the migration strategy from manual to automated work systems. What are the key considerations during this transition?	08	M	2	1
17	If the normal time is 1.30 min for a repetitive task that produces one work unit per cycle, and the company uses a PFD allowance factor of 12%, determine: (a) The standard time for the task and (b) How many work units are produced in an 8-hour shift at standard performance?	8	L	2	3



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18	The normal time for a repetitive task that produces two work units per cycle is 3.0 min. The plant uses a PFD allowance factor of 15%. Determine (a) The standard time per piece and (b) How many work units are produced in an 8-hour shift at standard performance?	8	L	2	3
19	The normal time to perform a certain manual work cycle is 3.47 min. In addition, an irregular work element whose normal time is 3.70 min must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%. Determine (a) The standard time per piece and (b) How many work units are produced during an 8-hour shift at 100% performance, and the worker works exactly 7.018 hr, which corresponds to the 14% allowance factor. (c) If the worker's pace is 120% and he works 7.2 hours during the regular shift, how many units are produced?	8	M	3	3
20	The normal time to perform a repetitive manual assembly task is 4.25 min. In addition, an irregular work element whose normal time is 1.75 min must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance factor is 16%. Determine (a) The standard time per piece and (b) How many work units are produced in an 8-hour shift at standard performance? (c) Determine the anticipated amount of time worked and the amount of time lost per 8-hour shift that corresponds to the PFD allowance factor of 16%.	8	M	3	3
21	The standard time for a manual material-handling work cycle is 2.58 min per piece. The PFD allowance factor used to set the standard was 13%. During a particular 8-hour shift of interest, it is known that the worker lost a total of 53 min due to personal time, rest breaks, and delays. On that same day, the worker completed 214 work units. Determine (a) The number of standard hours accomplished, (b) Worker efficiency, and (c) The worker's performance level expressed as a percentage.	8	M	3	3
22	The normal time of the work cycle in a worker-machine system is 5.39 min. The operator-controlled portion of the cycle is 0.84 min. One work unit is produced each cycle. The machine cycle time is constant. (a) Using a	8	M	3	3



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	PFD allowance factor of 16% and a machine allowance factor of 30%, determine the standard time for the work cycle. (b) If a worker assigned to this task completes 85 units during an 8-hour shift, what is the worker's efficiency? (c) If it is known that a total of 42 min was lost during the 8-hour clock time due to personal needs and delays, what was the worker's performance on the portion of the cycle he controlled?				
23	A worker is responsible for loading and unloading a production machine. The load/unload elements in the repetitive work cycle have a normal time of only 24 sec, and the machine cycle time is 2.83 min. One part is produced each cycle. Every sixth cycle, the operator must replace the tote pans of parts, which takes 2.40 min (normal time). For setting the standard time, the PFD allowance factor is 15% and the machine allowance factor is 15%. Determine the standard time under the following alternative assumptions: (a) the irregular element is performed as an external element and (b) the irregular element is performed as an internal element. (c) Determine the corresponding standard daily production quantities (8-hour shift) for each of these time standards.	8	M	4	3
24	The work cycle in a worker-machine system consists of (1) external manual work elements with a total normal time of 0.42 min, (2) a machine cycle with a machine time of 1.12 min, and (3) internal manual elements with a total normal time of 1.04 min. (a) Determine the standard time for the cycle, using a PFD allowance factor of 15% and a machine allowance factor of 30%. (b) How many work units are produced daily (8-hour shift) at standard performance?	8	M	3	1
25	The normal time for a work cycle in a worker-machine system is 6.27 min. For setting the standard time, the PFD allowance factor is 12% and the machine allowance factor is 25%. The work cycle includes manual elements totaling a normal time of 5.92 min, all but 0.65min of which are performed as internal elements. Determine (a) the standard time for the cycle and (b) the daily output at standard performance. (c) During an 8-hour shift, the worker lost 39 min due to personal time, rest breaks, and delays, and she produced 72 pieces. What was the worker's pace on the operator-controlled portion of the shift?	8	M	4	3